

Homework 1 – Marine Geology

Total 50 points

Due date: Tuesday 10/07/2025 at 10:45 am (before I leave the classroom)

Rules:

- **Group assignment: 3-4 students**
- **Individual assignments will not be accepted**
- **All answers must be handwritten. Typed and digital assignments will not be accepted**
- **Extensions will not be granted, except under unusual circumstances or examples covered in the section on Missed Class Policy and University Sanctioned Absences on Syllabus.**

- 1- Explain why stars are important to the ocean's formation, composition, and characteristics (10 Pts).
- 2- Read Elkins-Tanton (2013) article and answer: If both Earth and Venus were probably built from similar rocky materials and formed by similar mass-accretion processes, why is it that Earth has a marine oceanic ecosystem and viable atmosphere whereas Venus is lacking both features? (10 Pts)
- 3- Describe the theory of plate tectonics and how plate movement has reshaped the planet's surface over time. (10 pts)
- 4- Describe the various zones that the seabed is generally divided into and how scientists study the ocean floor. (10 pts)
- 5- Read Heffernan (2019) article and answer (10 pts):
 - a. Even though seabed mining may be economically lucrative, what are the biological concerns related to such largescale disturbance of the seafloor? (2.5 pts)
 - b. What type of sediments are the deep-sea miners interested in and why? (2.5 pts)
 - c. Are mid-ocean ridges the only deep-ocean feature where sulfide deposits form? Based on the Plate Tectonics class, where can we expect such minerals to form? (2.5 pts)

- d. Why we rarely find nodules and crusts on the continental shelf areas? Why? (2.5 pts)

6- **[MANDATORY] Group Accountability Exercise: In a short paragraph or a statement, describe the role of each group member during the homework assignment.**

ANY work done will be rewarded with a grade. Students with larger workloads will be rewarded extra points. Students with **NO CONTRIBUTION** to the group activity won't receive any points.

Example 1: All group members equally contributed to the homework assignment.

Example 2: Jane Doe answered questions 1 and 2, while John Doe, Jim Doe, and Carry Doe equally contributed to questions 3, 4, and 5.

Example 3: Everyone contributed except John Doe.

In example 2 Jane Doe will receive extra 5 points. In example 3, John Doe will fail the assignment (0 pts).

**IF THE GROUP FAILS TO SOLVE THE GROUP ACCOUNTABILITY EXERCISE,
THE ENTIRE GROUP WILL RECEIVE A TOTAL OF 0 POINTS ON THE ENTIRE
HOMEWORK ASSIGNMENT**